Lab 4 Case Study: Analysis, Synthesis, and the Complex Plane

Milestone: Complete at least 7 out of 14 of the questions to receive credit for in-lab.

Overview of the Lab

In this lab we analyze an unknown periodic signal, $x(t)$. Through this analysis you will uncover the embedded sinusoidal waves (their frequencies and phases) within this unknown signal. Using this information, you will test whether you can reconstruct the original signal. This requires moving from the time domain to the frequency domain, and then back to the time domain. In addition, you will reference how this all relates to the complex plane.

Goals

The goal of the lab is for you to solidify the concepts of:

1. How to decompose a signal into its sinusoidal components using a Fourier series.
2. Understand what information is held within a Fourier coefficient and its relationship to the time domain, frequency response, and complex plane.
   - Understand the relationship between the complex plane and the time domain.
   - Understand the relationship between the frequency response and the time domain.
   - Understand the relationship between the complex plane and the frequency response.
3. Reconstruct the signal in the time domain using Fourier coefficients, the information in the complex plane (e.g., unit circle), or the frequency response.

Why Are We Doing This?

This lab is the foundation for signal analysis. One of the hardest hurdles within signal processing is understanding the abstract connections among the complex plane, the frequency domain, and the time domain.

Entering the Unknown

Assume a fictional story: you are a subject-matter expert in signal processing contracted to analyze the structural integrity of a building. The structural engineer has time-series traces of how the building vibrates when perturbations are induced by high winds, large loads, etc. Because specific vibrations and their magnitudes determine how the structure is displaced as a function of time, this information indicates whether the building is safe. They state that any sinusoidal waveform over 30 Hz with amplitude greater than 8 could compromise the building. The firm wants to know whether specific frequencies exist in this vibrational data, as well as their magnitudes and phases (phasor information).

1 Working within the Time Domain

The firm sends a function `BuildingTraceData` which, for security reasons, must be supplied with your UFID (no dashes or spaces). When you do this, it returns two variables T_s and Y_t. The vector T_s contains the time samples; the vector Y_t contains the signal amplitude at those times.

1. Plot the signal in the time domain.
2. Upon plotting, you should see an unusual signal. Is the signal periodic? Why?
3. By visual inspection of the time series, measure the fundamental period T_0 (as discussed in class).
4. Using the fundamental period, calculate the fundamental frequency f_0. *Note:* the answer will not be exact; estimate to the nearest integer. Once f_0 is rounded, recalculate T_0 to reflect any minor change.

2 Fourier Series (Analysis)

As a signal-processing SME, you know you can decompose the unknown signal using a Fourier series. Although not always the best approach, it is the most fundamental for the firm to understand.

The Fourier series coefficient is defined as

$$A_k = \frac{1}{T_0} \int_0^{T_0} x(t)\, e^{-j2\pi kt/T_0}\, dt. \tag{1}$$

Here A_k is the Fourier coefficient, T_0 is the fundamental period, k is the kth harmonic, and t is time.

We cannot directly take the integral using complex exponentials in MATLAB unless the function is fully defined; doing the integral by hand would be impractical for non-trivial data. Instead, integrate the real and imaginary parts independently and then combine them to form a complex number, using Euler's formula

$$e^{j\theta} = \cos\theta + j\sin\theta. \tag{2}$$

Substituting into (1) gives

$$A_k = \frac{1}{T_0} \int_0^{T_0} x(t) \big[\cos(-2\pi kt/T_0) + j\sin(-2\pi kt/T_0) \big]\, dt. \tag{3}$$

Using identities

$$\cos(x) = \cos(-x), \qquad \sin(-x) = -\sin(x), \tag{4–5}$$

we rewrite

$$A_k = \frac{1}{T_0} \int_0^{T_0} x(t) \big[\cos(2\pi kt/T_0) - j \sin(2\pi kt/T_0)\big]\, dt. \tag{6}$$

Split the integral:

$$A_k = \underbrace{\frac{1}{T_0} \int_0^{T_0} x(t) \cos(2\pi kt/T_0)\, dt}_{A_k^{\mathrm{R}}} - j \underbrace{\frac{1}{T_0} \int_0^{T_0} x(t) \sin(2\pi kt/T_0)\, dt}_{A_k^{\mathrm{I}}}. \tag{7–9}$$

Thus

$$A_k = A_k^{\mathrm{R}} - j A_k^{\mathrm{I}}. \tag{10}$$

Perform the following:

5. Determine the first 10 *pairs* of Fourier coefficients A_k (complex numbers), from A_{-10} to A_{10}.
6. For each phasor term, determine the cyclic frequency f component and create a vector of those frequencies.
7. Plot the magnitude spectrum $|A_k|$ versus the corresponding cyclic frequency f.
8. Plot the phase spectrum $\angle A_k$ versus the corresponding cyclic frequency f.
9. From the frequency spectrum, determine the frequency of each non-zero phasor.
10. How much faster is the nth harmonic moving compared to the first harmonic? Give specific examples from your data to show this relationship for any n.

3 Fourier Series (Synthesis)

To double-check your work, reconstruct the signal using the synthesis equation

$$x(t) = \sum_{k=-\infty}^{\infty} A_k\, e^{j2\pi kt/T_0}. \tag{11}$$

Then:

11. Use the synthesis equation to reconstruct the received data based on your Fourier coefficients.
12. Were you able to properly reconstruct the signal? Plot the original and reconstructed signals together on one figure and compare with the original plot.
13. If the error is large or small, why? Can you improve the reconstruction? If so, do so and explain.
14. Finally, is the building safe? Explain using your data. If you improved the reconstruction, explain what you did and why it worked.